PHYSICAL SCIENCES

Designing microplastic-binding peptides with a variational quantum circuit–based hybrid quantum-classical approach

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De novo peptide design exhibits great potential in materials engineering, particularly for the use of plastic-binding peptides to help remediate microplastic pollution. There are no known peptide binders for many plastics—a gap that can be filled with de novo design. Current computational methods for peptide design exhibit limitations in sampling and scaling that could be addressed with quantum computing. Hybrid quantum-classical methods can leverage complementary strengths of near-term quantum algorithms and classical techniques for complex tasks like peptide design. This work introduces a hybrid quantum-classical generative framework for designing plastic-binding peptides combining variational quantum circuits with a variational autoencoder network. We demonstrate the framework's effectiveness in generating peptide candidates, evaluate its efficiency for property-oriented design, and validate the candidates with molecular dynamics simulations. This quantum computing–based approach could accelerate the development of biomolecular tools for environmental and biomedical applications while advancing the study of biomolecular systems through quantum technologies.

INTRODUCTION

De novo peptide design holds great promise for use in a wide array of applications ranging from therapeutics (*1*) and drug discovery (*2*) to synthetic biology (*3*) and material engineering (*4*). The design of function-specific peptides can enable the development of tailored peptide-based systems capable of interacting with biological (*5*), chemical (*6*), or inorganic targets (*7*) while providing insights into their underlying mechanisms. An interesting and unexplored area is the use of inorganic-binding peptides to remediate microplastic pollution (*8*). Microplastic contamination in various ecosystems poses serious environmental and health concerns (*9*, *10*). The ability of peptides to adsorb strongly to a variety of inorganic materials can be harnessed to help detect, capture, or facilitate biodegradation of microplastic waste. However, few or no peptides are known to bind to plastics, and collecting experimental data is time consuming and expensive. De novo computational peptide design can discover such peptides and thereby realize these microplastic remediation strategies. Conventional property-oriented de novo design approaches for peptides include rational design wherein a reference peptide sequence is modified by using chemical or physical intuition, followed by evaluation of peptide affinity using docking or molecular dynamics (MD) simulation (*11*, *12*). However, such an approach requires a reference peptide, which does not exist for many plastics, and input from experts. Alternatively, Monte Carlo methods, which do not require input domain expertise, can be used to either modify a reference peptide to optimize binding energy (*13*) or sample peptide sequences to see if they fit a certain structure (*14*), binding site (*15*), or material surface (*16*). Monte Carlo methods have been used to design peptides that bind to plastics like polystyrene, polypropylene, and polyethylene terephthalate (PET) (*17*), but they sample a small fraction of the large design space, which grows exponentially with peptide length. Data-driven approaches and alternative computational paradigms show promise in overcoming the computational challenges of these design methods.

A variety of generative deep learning approaches have been applied for de novo peptide design (*18*). Deep generative models like generative adversarial networks (GANs) (*19*, *20*) have been used primarily to design peptides tailored for specific applications, like antimicrobial peptides (*21*). Long-short term memory (LSTM) networks (*22*, *23*) and variational autoencoder (VAE) (*24*, *25*) models have been developed for generating bioactive peptides with antimicrobial and therapeutic properties (*26*). Reinforcement learning has been used to facilitate exploration during the generation of binding peptides (*27*). Ensemble learning techniques can offer improved exploration of chemical space relative to alternative deep learning methods (*28*), but at the cost of larger and possibly prohibitive amounts of training data for efficient learning. Deep learning methods inherently use large corpora of experimental data, such as those available for antimicrobial (*29*), anticancer (*30*), and cell-penetrating peptides (*31*). Despite their potential, such deep generative models have limited applicability for designing plastic-binding peptides due to the lack of large experimental datasets on peptide adsorption to plastics. When experimental datasets are sparse, one approach is to integrate a small, quantitative experimental dataset with computational modeling. For example, MD simulations and experimental measurements of peptide affinity were used to design peptides that bind selectively to either gold or silver (*32*), and to identify properties that give peptides high affinity for iron oxide (*33*). As these experimental and computational methodologies were tailored to the specific materials of interest, it is difficult to transfer them to other materials. For plastic-binding peptides, it is also costly and challenging to collect quantitative experimental data on peptide adsorption to plastics (*34*). While deep generative models excel in generating

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high-dimensional data efficiently, scaling these models to handle the vast search space of peptides (*35*) and ensuring their effective exploration of the huge space of peptide sequences and properties (*18*) poses a significant challenge. Current deep generative models also struggle to capture shallow statistical correlations in the data, which may limit their ability to explain underlying mechanisms in peptide design.

Quantum computing can help solve large-scale, combinatorial computational problems like de novo peptide design, owing to its speedup over classical computers (*36*). This speedup is crucial for the complex tasks of peptide design (*37*) for which the computational complexity can scale as N^{20} for a peptide with N amino acids. Quantum computing can manage this complexity more effectively by leveraging quantum properties like superposition and entanglement to facilitate efficient exploration of multiple configurations (*38*). This scalability is vital for peptide design. Additionally, quantum machine learning can utilize quantum circuits to extract features from high-dimensional data like peptide sequences (*39*) and potentially realize more accurate property predictions over classical techniques. Quantum algorithms have previously been leveraged for peptide and protein design challenges (*40*) with the aim of accelerating tasks such as searching low-energy configurations (*38*) or conformational sampling (*41*). Computational advantages offered by quantum computing can help address some of the challenges of classical deep generative models by enabling parameter-efficient quantum counterparts. Quantum generative models like quantum LSTM can handle sequences (*42*), while GANs can be implemented with quantum circuits to learn complex data distributions (*43*). Quantum VAEs have been shown to compress high-dimensional data into low-dimensional latent representations using quantum circuits; they have the potential to be scaled for large problems like peptide design (*44*). Implementation of these quantum algorithms on noisy intermediate-scale quantum (NISQ) devices (*45*) cannot, however, guarantee performance improvements for large-scale problems (*46*, *47*) due to the high-depth quantum circuit requirements for handling high-dimensional input data (*48*, *49*). Through integration of classical and quantum subroutines, data-driven methods can be paired with quantum computing to learn the structure-property relationships of peptides and guide the property-oriented design process. Hybrid quantum-classical approaches have been demonstrated to efficiently handle design tasks involving small molecules (*50*, *51*).

Here, we present a computational framework based on hybrid quantum-classical generative learning to design plastic-binding peptides by leveraging a near-term quantum computing technique, the variational quantum circuit (VQC). The VQC is integrated with a VAE network to predict the binding affinity of the peptides to the plastic PET and to represent the chemical space of peptides. This integration reduces constraints on circuit depth posed by gate-based quantum devices along with their associated noisy nature. The hybrid architecture is trained in an end-to-end manner, which allows exploration using the simplified peptide representation from the VAE and the expressivity offered by VQC. Gradient descent–based optimization is performed within the latent space by leveraging the trained networks to explore peptide chemical space that are then decoded into sequences with a high predicted affinity for PET. Atomistic MD simulations are performed and support the high affinity of the discovered peptides. A detailed analysis is presented on the applicability of the hybrid quantum-classical generative framework for designing plastic-binding peptides and its performance efficiency for optimizing peptide affinity. Lastly, the computational framework provided by near-term quantum computing technologies for a large-scale design problem is presented. The main contributions of this work are summarized as follows: (i) development of a quantum computing–based framework, integrating near-term quantum circuits with classical deep learning, trained in an end-to-end manner to both predict peptide affinity to PET and efficiently capture complex chemical features of the peptide; (ii) generative design of peptides with high binding affinity to PET that exploit the expressiveness of VQCs and the sampling capabilities of VAE to optimize peptide affinity within the latent space; and (iii) detailed analysis of the framework's ability and efficiency in generating plastic-binding peptides, including evaluation against conventional classical approaches.

RESULTS

Generative quantum-classical framework for peptide design

The proposed framework utilizes a dual-component system that leverages the complementary strengths of classical and quantum computing. The classical component is a VAE that encodes peptides consisting of 12 amino acids from one-hot encoded vectors into a compressed 16-dimensional (16D) latent space, as seen in Fig. 1 (A and B). The intermediate latent representations are the VQC inputs, where the quantum data embedding process encodes the latent vectors of the VAE into quantum states. These quantum states are further manipulated through parametric gate operations defined by the quantum ansatz in the VQC, followed by a measurement of the resulting quantum state that converts quantum information into classical information. A classical postprocessing step, shown in Fig. 1C, yields the predicted affinity of the peptide for PET. The predicted affinity is given by the PepBD score (*17*), a sum of the peptide-plastic interaction energy and the weighted peptide internal energy. All of the energies are composed of electrostatic, Lennard Jones, and generalized Born solvent energies. A lower score indicates stronger binding affinity. The hybrid quantum-classical framework is trained on a comprehensive dataset comprising over 350,000 peptide sequences previously evaluated by PepBD (*17*), each annotated with an affinity score that indicates the predicted binding affinity to PET. A joint training strategy is used to generate a structured latent space that can be more effectively processed by the VQC. To generate peptides, an initial latent vector is obtained with the encoder that is then optimized via gradient descent in the latent space using the trained VQC architecture, as shown in Fig. 1D. The peptide sequence corresponding to the optimized latent vector is obtained by using the trained VAE decoder to estimate the likelihood of each amino acid at every position in the sequence.

Latent representations lead to accurate affinity predictions of peptides to PET

The latent space representation separates peptides with high and low affinity into different regions, enabling the VQC to provide accurate score predictions. The t-distributed stochastic neighbor (t-SNE) embeddings for the latent representations and their affinity scores are visualized in Fig. 2. Notably, low-scoring and high-scoring sequences cluster into distinct regions that are separated by a transitional zone. This spatial distribution enables the VQC to effectively predict the score of a given sequence from the latent space.

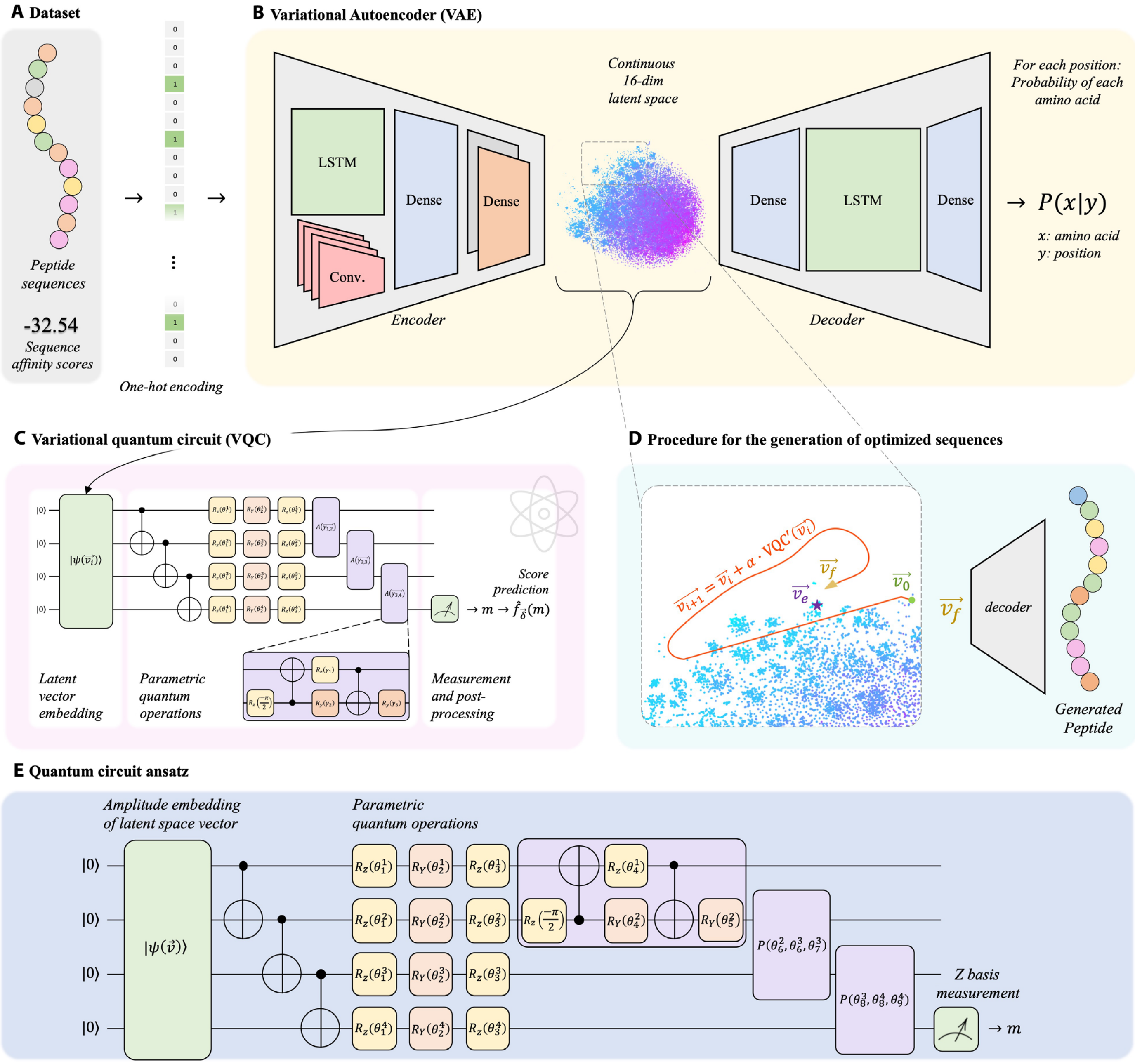

Fig. 1. An overview of the VQC-based peptide generation framework. (**A**) The training dataset consists of 12–amino acid peptides, each associated with a binding affinity score. These peptides are translated into one-hot encoded vectors and subsequently fed into a (**B**) variational autoencoder (VAE). The encoder of the VAE learns to compress the peptide information into a 16-dimensional continuous latent space, while the decoder learns to reconstruct the peptide sequence and estimate the conditional probability $P(x \mid y)$ for each amino acid's presence (x) at a specific position (y) within the sequence. (**C**) The captured latent space is also influenced by the variational quantum circuit (VQC), which learns to predict the corresponding affinity scores. The VQC and the VAE are trained in an end-to-end manner. (**D**) Optimization trajectory within the capture latent space to generate a novel peptide with a low affinity score is depicted wherein an initial latent vector, $\vec{v_0}$, is optimized iteratively using gradient descent to minimize the score predicted by the VQC. The optimal vector, $\vec{v_f}$, is subsequently decoded to synthesize a novel peptide. (**E**) The quantum circuit ansatz comprising single-qubit and two-qubit parameterized gates used in the VQC.

Particularly noteworthy are the sparsely populated areas within the regions associated with higher scores, as illustrated in Fig. 2C. These are potentially fruitful areas in which to discover novel peptides with high affinity for plastic PET. We also visualize the accuracy of the VQC in predicting the scores of peptides in the testing set in Fig. 2D. The predictive accuracy achieved through the joint training of the proposed hybrid quantum-classical framework is underscored by a Pearson correlation coefficient of 0.988 and a mean square error (MSE) of 4.754.

The importance of end-to-end training is demonstrated by comparison to a modular counterpart wherein the VAE and VQC networks are trained in separate sequential phases. The resulting t-SNE

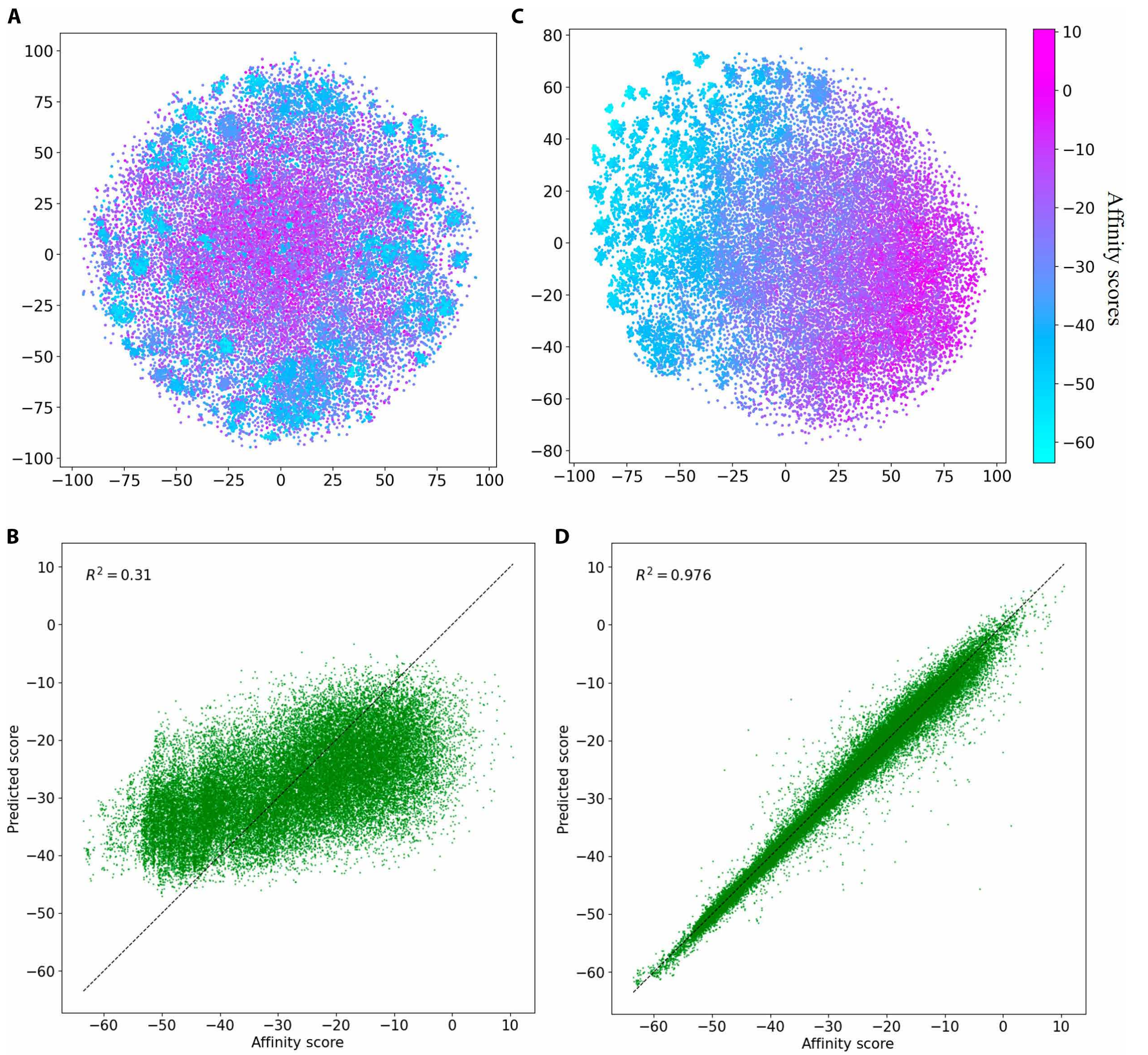

Fig. 2. Comparative analysis of sequential versus end-to-end model training. (**A** and **B**) Initial training strategy where the VAE and the VQC are trained sequentially. In (A), we observe the t-SNE visualization of the latent space after VAE training, with the color gradient indicating the binding affinity scores. Although some degree of clustering is visible, this sequential training methodology resulted in suboptimal VQC performance, as it struggled to accurately learn to predict the score. This is evidenced by the scatterplot in (B), where the predicted scores diverge significantly from the true scores (ideal alignment along the dashed line). Conversely, (**C**) and (**D**) present the results from the end-to-end training of the VAE and VQC. Plot (C) reveals a spatial distribution within the latent space that aligns with the scores. In contrast to (A), the VAE not only serves the purpose of data compression and reconstruction but also assists the VQC in score prediction. The efficacy of this co-training is corroborated by the scatterplot in (D), where there is an almost perfect alignment between predicted and true scores.

embeddings for the latent representations are presented in Fig. 2A, and the accuracy of VQC in predicting affinity scores is shown in Fig. 2B. While the VAE encoded peptide sequences into a homogeneous and continuous latent space, the VQC struggled to learn from the encoded sequences and poorly predicted affinity scores. Increasing the quantum circuit's complexity, either by incorporating an additional qubit for encoding the norm of the latent space vector or by augmenting the number of ansatz layers, resulted in only marginal improvements in the VQC's predictive accuracy. Thus, the poor score prediction was not improved by increasing the quantum circuit's expressiveness, even with latent representations effectively configured for peptide reconstruction. This can be attributed to the uniform dispersion of sequences from the training dataset, where their position is not correlated with their respective scores. In contrast, end-to-end joint training of the VAE and the VQC produces accurate affinity score predictions, and the latent space exhibits localization of low and high-affinity peptides, as illustrated in Fig. 2D. This observation underscores a strategic pivot in the focus of encoding: sequential training prioritizes a comprehensive feature extraction from peptide sequences, while joint training with the VQC balances feature extraction and score prediction. Additionally, joint training enables the VAE to adapt its representation of sequences to better suit VQC processing requirements. The cooperativity achieved by simultaneously training the classical VAE architecture and the quantum circuit for efficient affinity score prediction offers a robust framework for generating peptides with high affinity to PET.

Generative performance with VQC-based optimization

The trained classical and quantum components of the framework collectively can be used to generate peptides that have a high predicted affinity for PET. As shown in Fig. 1D, generating new peptide sequences begins by optimizing a vector in the latent space via gradient descent from an initial latent vector $\overrightarrow{v_0}$ to the final latent vector $\overrightarrow{v_f}$ to minimize the score determined by the VQC. Subsequently, the decoder processes the optimized vector to generate a new peptide sequence. Optimization performed with the VQC within the latent space is not constrained to a specific region of the latent space and enables exploration of the peptide chemical space along the optimization trajectory. The decoder translates the optimized latent vector into an actual peptide sequence. This is exemplified in Fig. 1D as the transition from the optimized latent space vector $\overrightarrow{v_f}$ to the actual decoded sequence $\overrightarrow{v_e}$. In the example in Fig. 1D, the decoder generated a sequence with a predicted affinity score of −61.46, whereas the optimized latent vector had a score of −62.23 kcal/mol. This variation is acceptably small compared to the variation in the scores in the training set, which range from −30 to −63.5. To achieve high accuracy in predicting scores for any peptide sequence, it is crucial to have a dataset with a broad spectrum of binding affinities. This is illustrated in Fig. 3C, which demonstrates that our dataset encompasses a wide range of affinities. However, it is important to note that our objective is to maximize predicted affinity for PET by minimizing the affinity score. Thus, we only want low affinity scores but seek diversity in the peptides with low affinity scores. The generated peptides have this property: The sequences of 12 best-scoring designs differ on average, in more than half of their residues with respect to any of the 12 best-scoring PepBD designs.

Using the abovementioned procedure generates novel peptide sequences with a high affinity for PET plastic. Affinity scores for the generated peptides are estimated by encoding the peptide followed

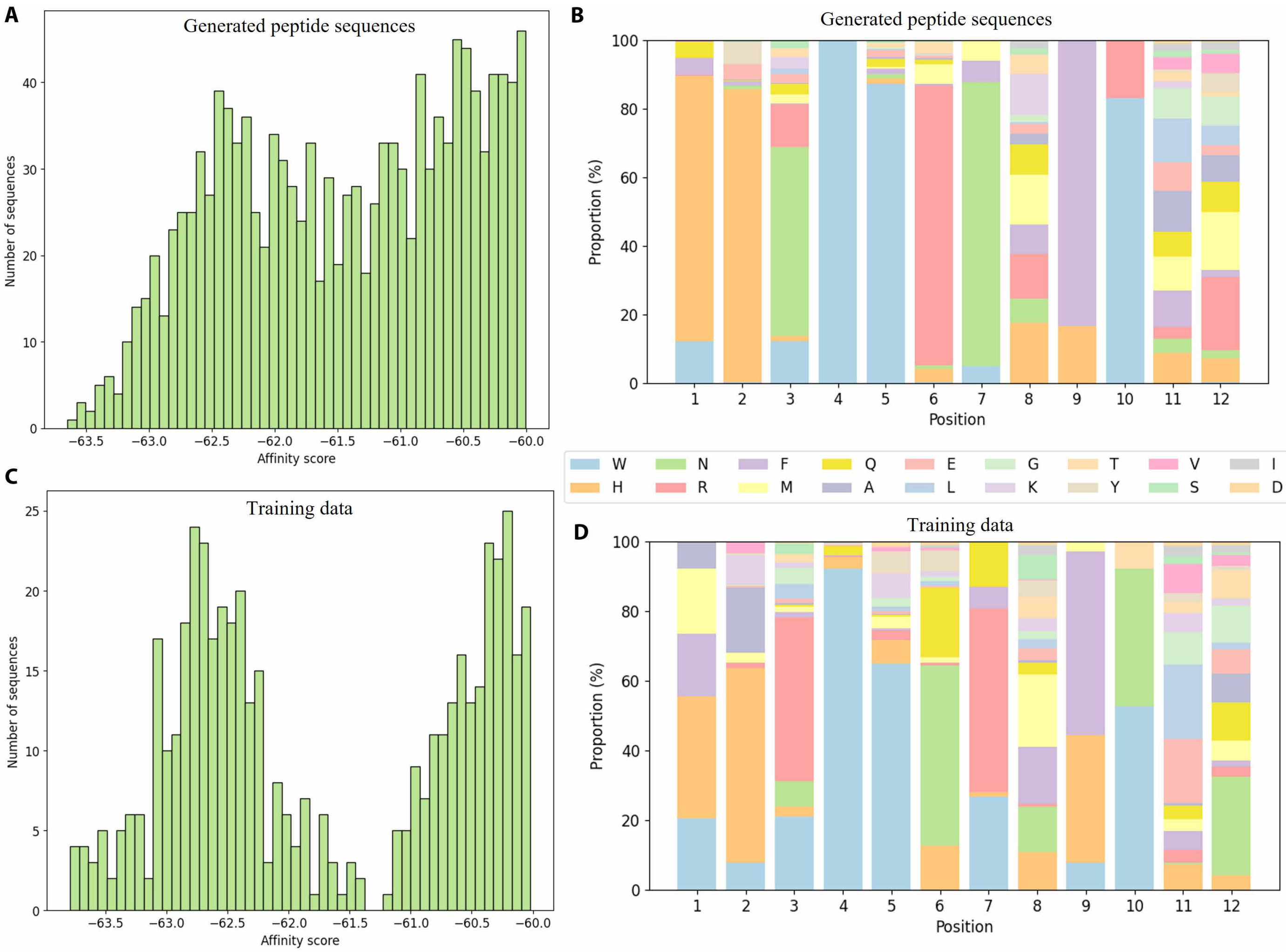

Fig. 3. Distribution of predicted scores and amino acid frequencies. The distribution of predicted scores for generated peptides with the VQC-based approach is shown in (**A**), while the observed distribution in the training set is presented in (**C**). The right panel illustrates the amino acid distribution at each position for the generated peptides in (**B**) and peptides in the training set in (**D**). Only peptides with observed and predicted scores lower than −60 kcal/mol are considered here.

by score prediction using the VQC. Score distributions of generated peptides are plotted in Fig. 3A, while the distribution of affinity scores for peptides in the training set is presented in Fig. 3C. The proposed VQC-based generative framework generates many peptide sequences with scores near to, or less than, −63.5 kcal/mol, the best score in the training set. As a lower score indicates higher predicted affinity, it appears that the model has expanded the high-affinity peptides beyond the initial dataset and can effectively generate PET-binding peptides. The amino acid distribution at individual positions along the peptide in the training set and the designs generated using the VQC-based approach are shown in Fig. 3B. The generated peptides generally show less variability than the training data, with several positions in the generated sequences predominantly occupied by certain amino acids. This likely results from the aim to optimize the affinity score, leading the framework to place greater emphasis on sequences in the training set with good affinity scores and to neglect sequences with poor scores. Overall, the VQC-based model generates peptides predicted to have a high affinity to PET.

Evaluating the suitability of designed peptides for microplastic remediation

The designed peptides were confirmed in MD simulations to have high affinity to PET relative to other peptides. MD simulations provide an essential validation of the accuracy of the affinities predicted by the VQC-based model. Peptides generated with three design approaches were evaluated: designs from the VQC-based model, PepBD designs (*17*), and randomly generated amino acid sequences. Twelve peptides of each type were evaluated, where the best-scoring sequences from PepBD and the VQC-based model were selected. Their amino acid sequences are provided in table S1. Comparison of the binding enthalpies and free energies of the peptides to PET, calculated using molecular mechanics generalized Born solvent area (MM/GBSA) method and normal mode analysis (*52*), shows that the VQC-based designs have comparable affinity with that of the best PepBD designs and greater affinity than random peptides, as illustrated in Fig. 4A. Thus, the high-affinity predictions from VQC are borne out in MD simulations. We also calculated the predicted water solubility of the generated peptides, a crucial property for

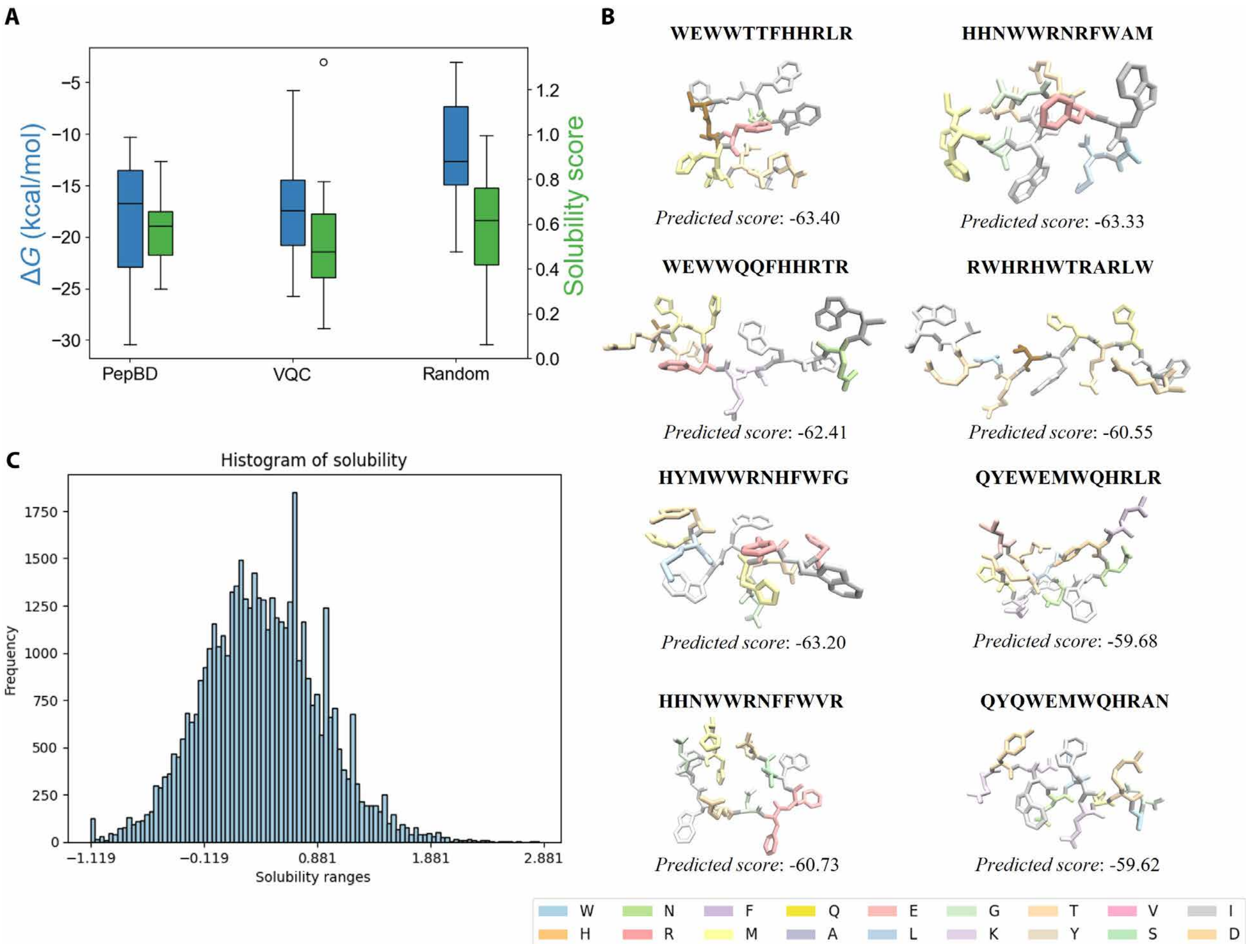

Fig. 4. Property evaluation for the generated peptides. (**A**) Binding free energies (Δ*G*) calculated using MM/GBSA strategy and solubility scores of peptides. Solubility scores were calculated using the CamSol method. The free energy values are obtained with MD simulations conducted with the peptide candidates and the binding plastic. A few peptide candidates generated with the VQC-based approach are shown in (**B**), while the distribution of their solubilities computed with CamSol exhibit a bell-shaped distribution in (**C**).

microplastic pollution remediation as aqueous environments are a common sink for microplastics. Peptide solubilities were predicted by calculating CamSol solubility scores (*53*), as shown in Fig. 4C. CamSol predicts a peptide's solubility in water based on its amino acid sequence, with a higher score indicating higher predicted solubility. The VQC-based designs have CamSol scores roughly equal to those of random sequences, an encouraging result that indicates that the VQC-based designs are not overly enriched in hydrophobic amino acids and should be at least moderately soluble in water. Together, the favorable affinity and water solubility of the generated peptides indicate that they could be effective for detecting or capturing PET microplastics.

The MD simulations of the generated peptides were analyzed to determine the origin of the high affinity that the VQC-based designs have for PET. The average interaction energies of each amino acid with the plastic, along with a decomposition of this energy into van der Waals (VDW), electrostatic, and solvation energies, are presented in Fig. 5 (B and C). Hydrophilic and charged amino acids either worsen or have no net contribution to the binding energy as their favorable VDW and electrostatic interactions wind up being cancelled out by unfavorable solvation energies. Bulky hydrophobic residues like tryptophan (W) and phenylalanine (F) improve the peptide's binding affinity since the VDW interactions are sufficiently strong to compensate for worsening of the solvation energy. Amino acids that contribute more to the binding energy generally occur with higher frequency, but notable exceptions are arginine (R), histidine (H), and glutamine (Q). These three hydrophilic residues weakly contribute to the binding energy, yet appear with high frequency in designs. This can be explained by noting that some peptide residues must be solvent exposed when the peptide is bound to PET. While solvent-exposed residues will not interact strongly with the plastic, hydrophilic residues like R, H, and Q are optimal in these positions since they interact favorably with water.

Advantages of the hybrid quantum-classical setting

The efficacy of the VQC in the proposed hybrid quantum-classical generative framework for peptide design was confirmed via an ablation study. Here, we constructed classical counterparts of the hybrid framework by integrating the VAE with classical neural network models to replace the VQC. The variants of classical neural networks include a feedforward neural net (MLP) and a convolutional neural network (Conv), which are constructed to mimic the functionality of the VQC within the proposed generative framework. The classical models were jointly trained by using a strategy analogous to that used for the VQC-based model. Upon completion of training, the model variants generated new peptide sequences using the same protocol as that of the VQC-based generative framework. The VQC-based generative model outperformed the classical models

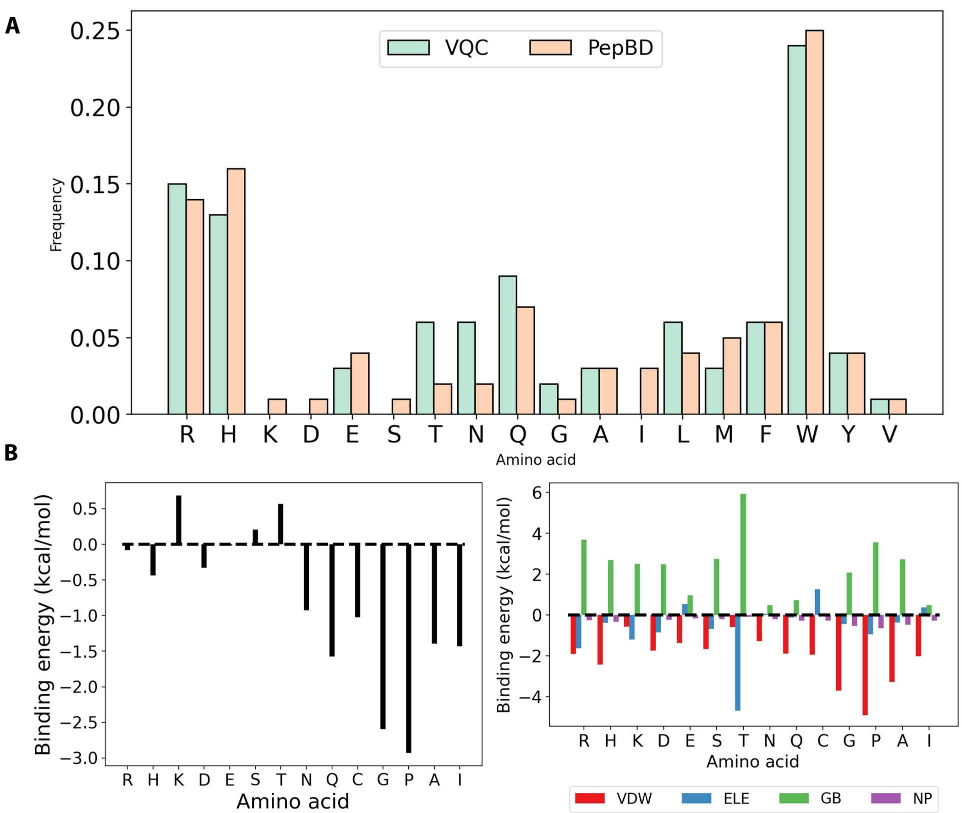

Fig. 5. Contribution of amino acid positions to the binding energy in the peptides designed with the proposed VQC-based approach and the PepBD algorithm. (**A**) Frequency of amino acids in the top 100 PepBD and VQC-based designs. (**B**) Average per-residue contribution of each amino acid type to the interaction energy (ΔH). Decomposition of per-residue contribution into van der Waals (VDW), electrostatics (ELE), generalized Born solvation energy (GB), and nonpolar solvation energy (NP) are also plotted on the right. For (B), amino acids are excluded if they were not present in any designs, and results are averaged over 96 simulations (8 simulations per peptide for 12 VQC designs).

in peptide generation, as shown in Fig. 6. To ensure a fair comparison between the methods, we applied the same optimization protocol across all models using identical sets of 900 initial latent vectors. Moreover, after sequence generation, all scores were predicted using an independently trained predictive model, ensuring uniform predictions and eliminating discrepancies. The VQC-based generative model is found to be superior to the other models. This is particularly evident in the peptide score distribution for VQC-based designs, which is skewed toward more negative values, indicating that the VQC model generates peptides with greater affinity toward PET plastics than the other models. This is highlighted in Fig. 6B, which shows the number of sequences generated by the different models with scores below −50 kcal/mol in intervals of 5 kcal/mol. The Conv-based model exhibits a score distribution skewed toward larger values than that of the original dataset, suggesting its inadequacy for this specific task of peptide sequence generation. The feedforward network displays a score distribution that is closer to the quantum model's performance and better than the dataset. However, it is still less effective than the VQC-based approach despite having more trainable parameters. The enhanced performance of the VQC-based model could be attributed to the VQC providing richer descriptions of the peptide with fewer parameters. Such advantages enable the VAE to encode sequences in an efficient manner, producing a score function landscape that aids gradient-based optimization and leads to high-quality local minima. An additional factor contributing to the VQC-based model's superior generative performance is its ability to more accurately predict affinity scores than the classical models, as demonstrated in Fig. 6C. The quantum-classical framework shows better scaling relative to classical alternatives for designing longer peptides. Figure 6C presents the number of trainable parameters within the property prediction component that estimates the binding affinity scores using VAE's latent representation as inputs, for each model used for benchmarking. This figure highlights that the VQC-based model has significantly fewer parameters than its classical counterparts of MLP and convolutional neural network. Additionally, the quantum circuit benefits from logarithmic scalability in relation to latent space dimension, unlike the classical models, which exhibit polynomial scalability. This feature is

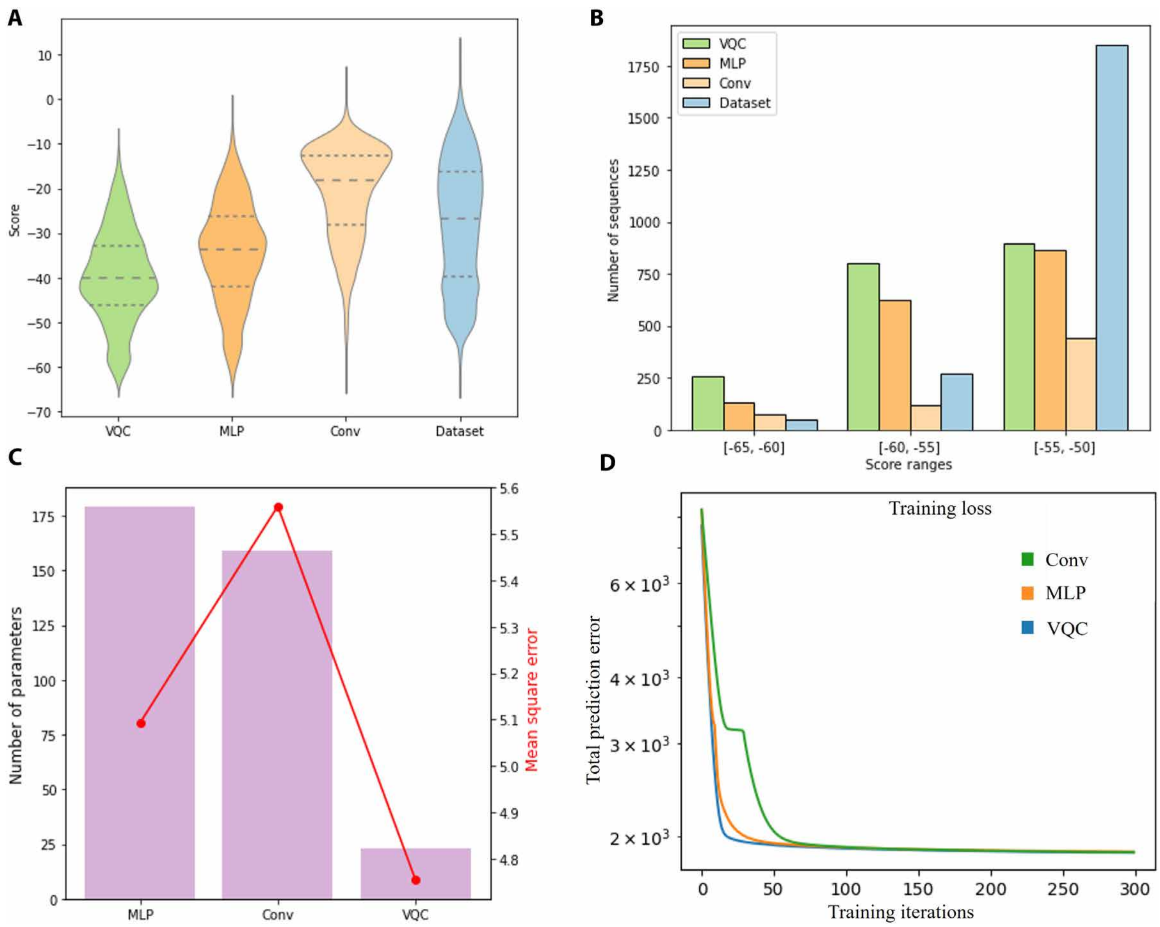

Fig. 6. Score distribution and comparative performance of sequence generation models. (**A**) Normalized distributions of predicted scores for peptides generated by the model with the VQC as the score predictor, compared to those produced where the VQC was replaced with either classical feedforward neural networks (MLP) or convolutional network (Conv), as well as the distribution of the training dataset. The quantum model demonstrates a better tendency to generate sequences with potentially higher binding affinities than the MLP or Conv models. (**B**) The total number of sequences generated within the lowest score ranges highlights the model's capability to explore regions of the score space not represented in the original dataset. All models used the same optimization protocol on 900 initial latent vectors to produce these results. (**C**) Comparison between predictive performance achieved with classical and VQC-based approaches. (**D**) Training loss curves are plotted for the proposed quantum circuit-based model and the selected baselines.

particularly promising for designing complex molecular structures that require large latent spaces.

DISCUSSION

The PET-binding peptides designed in this work can be a useful tool in the fight against microplastic pollution. Bio-based solutions hold much promise for mitigating PET plastic pollution given the optimization of enzymes that degrade PET (*54*) and the engineering of microorganisms to express these enzymes (*55*). The peptides designed in this work could facilitate remediation of PET pollution by facilitating degradation through incorporation into the PETase sequence (*8*) or facilitating microbial adhesion to PET waste by incorporating the peptide into the microbe's genome. These two steps can be readily achieved with current biotechnological tools. It should also be noted that the methodology can readily design peptide binders for other plastics for which modeling data are available, thereby expanding the toolkit of bio-based strategies for microplastic remediation.

The crux of the proposed approach lies in the innovative integration of a classical neural networks with a quantum circuits, harnessing their combined strengths in sequence encoding and score prediction to derive practical utility for property-oriented peptide design. This synergy has been pivotal in transcending the limitations observed in classical deep generative learning approaches investigated in this research. Similarly, quantum neural networks like quantum VAE can be directly implemented with quantum circuits (*44*), but can pose challenges in terms of scalability and sample complexity. The depth of the quantum circuit required to realize a quantum VAE can be prohibitive, and sampling such latent representations has high computational complexity (*49*). Although this work has yielded promising results using noisy state simulations of quantum circuits, it is important to note that the use of current quantum hardware might affect the outcomes, due to present technological limitations such as quantum noise and limited coherence, which can introduce significant errors in the calculations. Our use of a low-depth quantum circuit can help to mitigate such issues. However, these factors require careful evaluation and use of hardware-appropriate ansatz in the future to validate the actual capacity of current quantum computers in solving practical problems associated with peptide and protein design.

The quality of the peptides generated by the VQC-based framework depends on the diversity and quality of the training data. Greater diversity allows the framework to learn a more general relationship between a peptide sequence and affinity, while higher quality allows the model to more accurately predict peptide affinity for PET. We used a dataset generated by the computational modeling program PepBD algorithm (*17*). PepBD data are inherently diverse as they perform Monte Carlo sampling and use many different peptide structures adsorbed to PET to initiate peptide design, allowing PepBD to sample more than 400,000 unique amino acid sequences. On the other hand, the quality of PepBD data may be limited. Peptide affinity for PET was computed with the MM/GBSA method (*52*), which uses an implicit solvent model and does not account for the reduction in peptide configurational entropy upon adsorption. These two factors could influence the prediction of peptide relative affinity for PET (note that we are interested in relative peptide affinity, not absolute affinity). The use of a computational dataset is necessary because a quantitative dataset of peptide affinity for PET does not exist, and there are no high-throughput methods to generate such a dataset. This makes validation of our peptide designs essential. The simulation results in Fig. 4A are promising, but could be further validated using methods like quartz crystal microbalance or atomic force microscopy. Since such experiments can be time consuming and expensive, it may also be prudent to screen peptides by performing enhanced sampling MD simulations to identify the most promising peptides to test experimentally.

MATERIALS AND METHODS

Quantum-classical generative framework architecture

In this section, we introduce the architecture used by the quantum-classical generative framework to design peptides. The foundation of the hybrid quantum-classical architecture is a VAE that captures the latent chemical space spanned by the peptide dataset, and a VQC that provides affinity predictions and can be used for optimization within the latent space. The VAE comprises two components, namely, the encoder network to transform input peptide sequences into a continuous latent representation and the decoder network to reconstruct peptide sequences from their encoded latent vectors (*56*). Input to the encoder is a one-hot encoded representation of the peptide sequence indicating the amino acid at each position of the 12–amino acid chain constituting the peptide. This sequence is passed on to a long short-term memory (LSTM) network concurrently with a 2D convolutional (Conv) layer. This is followed by affine transformations with a fully connected network and further activated with a rectified linear unit (ReLu). Outputs of the encoder comprise a means vector and log variance denoted by $\vec{\mu}$ and $\log\left(\overrightarrow{\sigma^2}\right)$, respectively. During training, the latent vectors, $\vec{v} = \mathcal{N}\left(\vec{\mu}, \overrightarrow{\sigma^2}\right)$, introduced to the decoder consist of the calculated mean perturbed by Gaussian noise scaled by the variance, ensuring variability in the generation process. The architecture of the decoder consists of a dense layer that expands the latent space vector, followed by an LSTM network. A subsequent dense layer with a SoftMax activation function produces the probability distribution of amino acids at each position in the reconstructed peptide sequence. Figure S2 provides a comprehensive schematic of the VAE architecture.

The VQC is constructed to predict the affinity score from the encoded latent representation of the peptide sequence. The quantum circuit execution process initiates with quantum embedding to encode the classical information into a quantum state. We adopt amplitude encoding, which embeds a latent vector, $\vec{v_i}$, into an n-qubit quantum state represented as $|\psi(\vec{v_i})\rangle$. For a D-dimensional latent vector, $n \equiv \lceil \log_2 D \rceil$ qubits are required due to the logarithmic scaling facilitated by amplitude encoding. The initial quantum state is processed by layers of parametric quantum gates involving singe-qubit rotations, $R_z(\theta_i)$ and $R_y(\theta_i)$, and controlled NOT (CNOT) as entanglement gates. The parametric gate operations are dictated by the underlying quantum ansatz presented in fig. S3. A measurement operation at the terminal qubit is performed with the resulting quantum state and a Pauli-Z operator to convert quantum information into a classical scalar m. A postprocessing operation is further applied to yield the predicted affinity score, $\hat{y} = f_{vqc}(\vec{z}) = \delta_1 m + \delta_2$. The trainable parameters of the circuit include rotation angles $\vec{\theta}$ and $\vec{\gamma}$ along with the postprocessing coefficients $\vec{\delta}$, enabling the quantum circuit to be optimized for accurate peptide score estimation. Figure S3 illustrates the integration of VQC and VAE within the quantum-classical framework.

Data

A dataset of peptides with affinity for PET surfaces was created using the PepBD algorithm, which is a biophysics-based algorithm that designs peptides that bind to various plastics, such as PET (*17*). PepBD begins with a specific peptide sequence attached to a plastic surface. It then employs a Monte Carlo method integrated with simulated annealing to modify the amino acid sequence constituting the peptide. The optimization involves iterative Metropolis Monte Carlo steps where peptide mutations are accepted or rejected based on the resulting affinity score changes. For our experiments, data from 55 optimization trajectories have been compiled into a dataset comprising amino acid sequences along with their binding affinities to PET plastics. The constructed dataset contains 381,066 peptide sequences, each composed of 12 amino acids, and each peptide is annotated with a binding affinity score. The amino acids are represented by letters, meaning each sequence is a string of 12 letters. The dataset is split into a training set with 90% of the peptides and a test set with the remaining 10%. The training set is used to train the model, while the test set is used to evaluate the model's performance. The range of the affinity scores is between −64 and 12 kcal/mol, with a mean of −27.7 and an SD of 14.11. The distribution of the affinity scores in the dataset is shown in fig. S1. All peptides in the dataset have a fixed length of 12 residues. The peptide length is fixed in PepBD since the peptide score depends on the interaction energy with the plastic, which naturally increases as the peptide length increases. The result is that the peptide would grow to the maximum permitted length if the peptide length were allowed to vary. This is not a desirable feature, so the peptide length is fixed during peptide design. In choosing a peptide length, a short length (such as 12 residues) is useful because peptides can then be synthesized either chemically or biologically, thereby expanding accessibility to the peptides and the application of the peptides toward microplastic remediation. If peptides are chemically synthesized, the yield increases as the peptide length decreases since fewer reactions are needed. Fixing the peptide length specifically at 12 residues also facilitates comparison to peptides from experimental screening of peptide libraries, which are often of the same length. Thus, fixing the peptide length at 12 residues facilitates peptide design and increases the utility of peptides for microplastic remediation.

End-to-end training

Using the constructed dataset comprising peptides and their affinity scores, the VAE and VQC are trained in a joint manner, enabling the VAE to facilitate the score predictor's task by organizing the latent vectors in a manner that simplifies the VQC's task of processing information and making accurate predictions. The loss function used is a composite of the losses from both the VAE and the VQC-based score predictor. The VAE's loss is composed of the reconstruction loss, which is measured using cross-entropy to assess the accuracy of the input peptide reconstruction, and the Kullback-Leibler (KL) divergence as a regularization term, which encourages the latent space to approximate a multivariate Gaussian distribution. This combination ensures that the model not only accurately reconstructs the input data but also maintains a structured, continuous latent space. Such space will allow us to decode previously uncharted regions of the latent space to generate novel peptide sequences. The loss function for the VQC is the MSE between the predicted score $\hat{y}$ and the affinity score of the sequence y in the dataset. The operations associated with the VAE are conducted with a Nvidia GeForce RTX 3080 GPU, while the quantum circuit execution is carried out on an AMD Ryzen 3960X 24-Core CPU. The same CPU is also used to carry out all data preprocessing and postprocessing tasks. Additional details on implementation and training are presented in section S6.

Peptide generation

Our approach involves optimizing a vector in the latent space to minimize the score prediction with the quantum circuit to generate peptide candidates. The optimization task can be denoted as $\vec{z}* = \mathrm{argmin}_z f\left(\vec{z}\right)$, where $\vec{z}$ is a vector in the latent space. A gradient descent-based approach is adopted to perform optimization within the continuous latent space. Each gradient descent step can be denoted by $\vec{z}_{n+1} = \vec{z}_n - \eta \cdot \nabla f\left(\vec{z}_n\right)$, where η is the learning rate and the gradients $\nabla f\left(\vec{z}_n\right)$ for the VQC-based estimator are computed with the parameter shift rule (*57*). After optimizing a latent vector and decoding it, we obtain the conditional probability distribution representing the likelihood of each amino acid occurring at specific positions within the sequence. By sampling from this distribution, we can generate multiple distinct sequences, each with a unique affinity for the target surface. Once the model is fully trained, we can harness it to design new peptides with high affinity for PET. After optimizing, the optimized vector is decoded to identify the most probable corresponding amino acid sequence. Since the VQC was trained using samples from the normal distribution generated by the encoder for each peptide sequence, we expect that it will accurately predict the score for novel vectors in the latent space. Simultaneously, the VAE's decoder should be capable of pinpointing the amino acid sequence that either matches or closely resembles the optimized vector. This method effectively combines the predictive capabilities of the VQC with the decoding proficiency of the VAE, aiming for efficient generation of target-specific peptide sequences. This approach allows for the exploration of a diverse range of potential peptide sequences, each tailored to exhibit high binding affinity. By varying the optimization parameters like learning rates, we can navigate through different regions of the latent space, potentially uncovering new sequences with desirable properties. Additionally, the process of multiple samplings from the decoded distribution increases the likelihood of identifying amino acids that contribute to an optimal peptide structure, enhancing the overall efficacy of the peptide generation process. Figure S7 presents various optimization trajectories with different gradient descent schemes like stochastic gradient descent and Adam (*58*) with varying hyperparameters associated with peptide sequence generation.

MD simulations

MD simulations are used to evaluate the binding free energy of a peptide to PET plastics. While enhanced sampling methods like metadynamics (*59*) or umbrella sampling (*60*) can effectively calculate peptide adsorption free energies, their high computational cost makes them impractical for high-throughput peptide evaluation. We instead perform an ensemble of short equilibrium MD simulations for each peptide to sample many different adsorbed conformations and assign peptide affinity based on the most stable adsorbed conformation. This scheme is in contrast to a long equilibrium MD simulation, which is typically kinetically trapped in one conformation for the entire simulation. An initial ensemble of adsorbed structures for each peptide obtained by running a high-temperature (550 K) simulation that permits the peptide conformation to change rapidly, and then performing *k*-means clustering with CPPTRAJ (*61*) to obtain

16 representative conformations of the peptide adsorbed to the PET surface. Each of the 16 conformations was simulated for 1 ns before calculating the peptide's binding free energy with Amber's MMGBSA package (*62*). The eight simulations with the lowest binding free energy were extended an additional 4 ns before again calculating the binding free energy. The peptide's binding free energy was set equal to the lowest binding free energy from the eight extended simulations, as the associated peptide conformation is stable and assumed to be representative of the lowest energy absorbed structure.

The MD simulation details are as follows. The force fields used were TIP3P for water (*63*), GAFF (*64*) for plastics with partial charges taken from our previous work (*17*), and the ff14SB force field (*65*) for peptides. Atomistic models of plastic surfaces were taken from our previous work (*17*). The tLEaP40 module in Amber (*66*) first generated an extended, linear conformation of the peptide. The peptide and surface were then merged, and the peptide was translated and rotated so that its long axis was parallel to the plastic surface and there was a 4 Ā gap between the top of the surface and the peptide center of mass. tLEaP added TIP3P water 15 Å above the peptide and 10 Å below the bottom of the plastic surface. The simulation box dimensions parallel to the plastic surface were set equal to the dimension of the PET surface, which was 80 Ā in both directions. The Parmed utility of Amber (*62*) was used to translate coordinate and parameter files from Amber format to Gromacs format, and the simulations were run with Gromacs version 2019.6 (*67*). Before running high-temperature simulations at 550 K, the system was energy minimized using steepest descent for 1000 steps, heated in an NVT ensemble to 300 K over 100 ps, equilibrated in an NPT ensemble at 1 bar and 300 K for 200 ps, and heated in an NVT ensemble to 550 K ensemble for 200 ps. The 550 K simulation was finally run in the NVT ensemble at 550 K for 10 ns. After extracting 16 different conformations with *k*-means clustering, each conformation underwent an NVT simulation at 300 K for 100 ps to cool the system back to room temperature, followed by a production phase simulation in the NVT ensemble at 300 K for 1 ns. After calculating the binding free energies of all 16 simulations, the 8 simulations with the lowest binding free energy underwent an additional NVT simulation at 300 K for 4 ns. All carbons in the PET surface were restrained using position restraints with a force constant of 5000 kJ mol^{-1} nm^{-2}. Bonds to hydrogen were restrained using the LINCS algorithm (*68*). Long-range electrostatic interactions were treated using particle mesh Ewald. All simulations used a time step size of 2 fs. The thermostat in NVT and NPT simulations used the velocity rescaling algorithm (*69*) with a time constant of 0.1 ps. Two thermostats were used, one for all water molecules and one for all nonwater molecules. The semi-isotropic Berendsen thermostat (*70*) was used in NPT simulations to equilibrate the system density, where the *x-y* dimension changed independently from the *z* dimension. The barostat used a time constant of 5 ps and an isothermal compressibility of 4.5×10^{-4} for both the *z* and *xy* directions.

Statistical analysis
No statistical analyses were performed in this work.

Supplementary Materials
This PDF file includes:
Supplementary Text
Figs. S1 to S8
Tables S1 and S2
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Data and materials availability: All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. The peptide dataset used for training along with their properties is made available on Zenodo (doi.org/10.5281/zenodo.14014530) and GitHub (https://github.com/PEESEgroup/peptide-vqc). The PepBD code used for benchmarking is available in GitHub at https://github.com/CarolHall-NCSU-CBE/PepBD_Plastics.

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